

Phase of Data Gathering and Preprocessing

Date of Report	12 August 2026
Project Name	AI Career Pulse: Deciphering the Employment Market

Report on Data Quality

Prior to conducting analysis in Tableau, the AI job-market dataset underwent a quality review during the initial phase. The following data quality evaluation was executed on a total of 1500 records.

Source of Data	Identified Quality Issue	Severity Level	Proposed Resolution
AI Job-Market Dataset	Absent values: Across 25 fields within the AI Job Market Excel file, no cells were found to be empty.	Low	For this dataset, there is no need for data imputation or treatment of missing values.
AI Job-Market Dataset	Duplicate entries: No rows containing duplicate information were found, nor were there any duplicate entries for the 'job id' field.	Low	No duplicate records require deletion within this dataset.
AI Job-Market Dataset	Experience mismatch: A discrepancy exists between 'experience_level' and 'years_of_experience' in 1099 of the 1500 records. For instance, record AIJOB0002 indicates 2 years of experience but is categorized as 'Senior (6-9 yrs)'. 	High	Experience level should be derived from the 'years of experience' field. We must standardize the experience level indicator to match established experience brackets (e.g., Entry: 0-2, Mid: 3-5, Senior: 6-9, Lead: 10+). Furthermore, 'years of experience' might be necessary for standardizing related fields like the experience flag.
AI Job-Market Dataset	Salary bound inconsistencies: In 288 instances (19.2%), the 'annual salary usd' exceeds the 'salary max usd'. As an example, record AIJOB0003 shows an 'annual_salary usd' of \$360,000, whereas the 'salary max usd' is only \$300,000.	Moderate	We advise verifying the original definitions of 'annual salary usd' against 'salary min usd'/'salary max usd'. This will help determine if 'salary

			max usd' contains a typographical error. Any discrepancies should be resolved based on source logic if these fields are intended to represent the same salary bracket.
AI Job-Market Dataset	Salary bracket alignment: Thirty records possess 'annual salary usd' figures that conflict with their designated salary-tier descriptions. For example, some records with a \$100,000 annual salary are incorrectly tagged as 'Entry (<\$100k)'.	Moderate	It is advisable to establish uniform boundaries for salary tiers and recompute the salary tier based on the annual salary using a single, clearly documented guideline. This step should be completed before creating the Tableau dashboard.
AI Job-Market Dataset	Range verification: Demand score falls between 68-98, benefits score_10 ranges from 6.0-9.8, posting month is 1-12, and posting year is 2025-2026. No values breached these established ranges.	Low	No corrective actions are needed for range validation since every value stayed within the specified boundaries.

Domain of Cleansing	Execution Steps
Missing data	Examine all 25 fields for any empty entries that might necessitate imputation or further treatment. Currently, no missing data has been detected.
Duplicate records	Inspect all fields to locate duplicates. At this current stage, no redundant records were uncovered.
Experience parameters	Verify alignment between experience levels and years of experience, updating the experience_level field wherever necessary.

Salary attributes	Verify that the annual salary used values align properly with the minimum and maximum salary thresholds, clarifying underlying logic if required.
Salary classification	Recompute and standardize salary tiers using annual compensation in accordance with the dataset description.
Final verification	Re-examine all fields following revisions to confirm zero remaining data quality issues, ensuring the dataset is fully prepared for Tableau.